NELSON MANDELA UNIVERSITY

JUNE 2019 EXAMINATION

MODULE DESCRIPTION	:	Database Systems
MODULE CODE	:	WRDB301 / WRDV301
FACULTY	:	Science
QUALIFICATION	:	Various
EXAMINATION DATE	:	14 June 2019
SESSION STARTS AT	:	09:00
DURATION (IN MINUTES)	:	180
TOTAL MARKS	:	100
PAGES	:	8
ADDENDUMS	:	Appendix A (to be used in Question 5)
EXAMINER	:	1. Dr L Barnard
MODERATOR	:	1. Mr GJ Dollman (UFS)
INSTRUCTIONS	:	Follow the instructions on the page provided Answer all questions **Submit your answers in the folder provided**
REQUIREMENTS	:	Exam folder (no exam books required) Clean copy of *Database Principles* texts permitted (various editions from various authors available)

DO NOT TURN THE PAGE BEFORE BEING TOLD TO DO SO

Instructions for Question 1:

1. **The following diagram needs to be completed using MS Visio.**
2. **Select the SIMPLE theme in the DESIGN tab.**
3. **Save your diagram using your student number as filename in your assessment folder T:\WRDV301\<your username>.**
4. **Once you have completed your diagram,**
    a. **ensure your diagram fits onto a single A4 page (you may change the orientation - for this page only - to landscape, if required) and**
    b. **use the Snipping Tool to copy your diagram into your MS Word document.**
5. **Make sure that you have numbered your questions correctly in your MS Word document.**

1. Consider the following scenario. [29]

Animal Care Centre (ACC) runs a number of clinics where pet owners can take their pets to receive medical care. Every clinic has a unique clinic code (cCode) and a clinic name. Each of the clinics has a number of staff members that works at the specific clinic. Staff members do not rotate between clinics. Each clinic usually (but not always) has a single staff member that manages the clinic. For every staff member, the following details are recorded: unique staff member number (staffNo), staff member name, and staff member date of hire (the day on which the staff member started working at ACC). A staff member may be a medical staff member or an administrative staff member. For a medical staff member, the registration number and the date of registration at the veterinary council are recorded. For admin staff members, the job description is recorded.

The first time a pet owner contacts a clinic, the owner and the pet are registered at the clinic. All the ACC clinics has many registered pet owners. A pet has a single owner and the pet is always taken to the same clinic for examination and treatment. An owner can have many pets registered at the clinic. The following information is recorded on pet owners: unique owner number (ownNo), owner name, and owner contact number. A pet has a unique pet number (petNo). The name of the pet, the breed and date of birth of the pet are also recorded.

When the owner brings a pet to the clinic, a staff member will examine the pet. This examination may result in the pet being prescribed some treatment. In some cases, the pet is examined, but no treatment is necessary; in other cases, a pet might require multiple treatments. For each examination, the unique examination number (exNo) and the date of the examination is recorded. Each treatment has a unique treatment code (treatCode) and a description.

Make use of Visio and draw an implementation ready database design using Crow's Foot database notation.

Make sure you follow diagramming standards and ensure all attributes are atomic. Show at least the attributes as indicated in the scenario. You need to clearly indicate all primary keys and foreign keys, using appropriate notation, for all relations in your diagram. You may add additional attributes and surrogate primary keys, if required. Record any assumptions made, but ensure that the assumptions do not contradict the business rules provided in the scenario.

2. Examine the table given below. This table indicates the hours that temporary staff worked at each branch of a company and the maximum number of hours they may work, depending on their position. The table has a composite primary key, namely staffNo and brnchNo. [18]

staffNo	**brnchNo**	brnchAddress	staffName	staffPosition	hrsPerWk	maxHrs
15447	1001	16 Ring Rd, Greenacres	Dube, M	Salesrep	16	20
25485	1001	16 Ring Rd, Greenacres	Pikoli, T	Assistant	8	10
15447	1002	31 Drophy St, Bethelsdorp	Dube, M	Assistant	7	10
32894	1002	31 Drophy St, Bethelsdorp	Blatt, O	Runner	12	15
15447	1003	4 Mkhombe St, Motherwell	Dube, M	Salesrep	13	20
15489	1004	98 Bumbana St, Motherwell	Pikoli, TA	Assistant	9	10

a) The structure of this table make the data in the table susceptible to anomalies. Use the information provided in the table and explain how the following anomalies could occur: 4

- Insertion
- Deletion

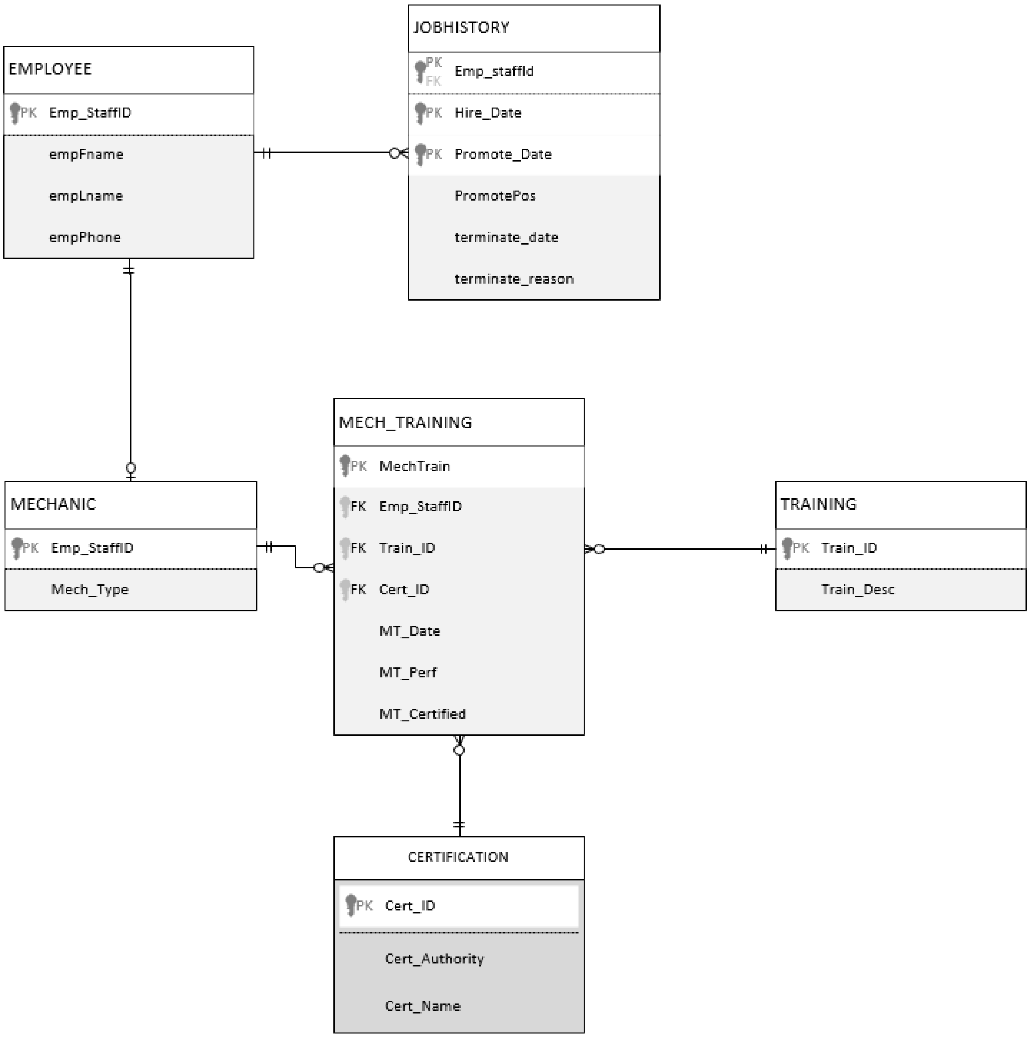
b) Identify the functional dependencies represented by the data provided in the table. Identify the desirable and all undesirable dependencies. For the undesirable dependencies you need to indicate the type of dependency. 8

State any assumptions you make when identifying the data. Give the functional dependencies in the appropriate format:

Determinant → Dependent

c) Use the functional dependencies identified in (b) and convert the given table into a set of third normal form (3NF) tables. Only give the relational schemas, and clearly identify all primary and foreign keys. Only use the attributes provided and do not make any changes to the attributes. 6

3. Consider the partial ERD below and then answer the questions that follow. [13]



a) Are the following entities strong or weak entities? Explain your answers in detail. 6

- JOBHISTORY
- MECH_TRAINING

b) Explain the type of relationship degrees present in the given diagram. 5

4. Consider the following sequence of actions (S1), listed in the order it is submitted to the DBMS, and followed by the commits of transactions that have completed. [10]

The timestamps of the transactions are as follows (with T2 being the oldest transaction and T5 being the youngest):

S1: T3:R(A), T5: R(A), T3: W(A), T2: W(B), T3: W(E), T4: W(E), T5: R(E), T3 commit

Make use of timestamping concurrency control, implemented using the Wound/Wait scheme, to show how the sequence is handled.

5. Use the Invoice and explanatory notes given in Appendix A to develop a conceptual model consisting of only the relational schemas that could be used to extract the data required to populate this form. You need to ensure the following: [10]
   - Only relational schema, normalised design to 3NF
   - Use appropriate, atomic attribute names
   - Indicate all primary and foreign keys

   Hint: some information displayed on the form would be derived attributes.

Instructions for Question 6:

1. **The following SQL queries need to be completed in MS Access.**
2. **Open Lecturer.accdb from your assessment folder (T:\WRDV301\<your username>).**
3. **Complete the queries using MS Access.**
4. **Save your queries, using the question number as query name, e.g. 5a or 7b etc.**
5. **Once you have completed the queries, copy the following to your MS Word document:**
   a. **your SQL code and**
   b. **make use of the Snipping Tool to copy a portion of your resultant table (which also shows the number of records in your resultant table – see example in Figure 1).**


5a

holder_id	Holder_FName
1	Fiona
2	Zizipho
5	Noel
9	Koos
10	Paseletso
11	Ignatious
12	Corne
13	Zebulon
14	Schalk
17	Omphile
19	Hennie
20	Annamarie

ord: 1 of 52

Figure 1 Sample resultant table

6. **DO NOT delete the database.**
7. **Make sure that you have numbered your questions correctly in your MS Word document.**

6. The relational schemas listed below show the tables defined in the lecturer database for storing information about lecturers, the modules offered, students, and the modules taken by students, and the lecturers supervising students. [14]

   The relational schema for the LECTURER.accdb is as follows:
   LECTURER(L_No, L_FName, L_LName, L_Office, DName, L_email)
   DEPARTMENT(DName, HoDLno, Building)
   STUDENT(Stu_No, Stu_FName, Stu_LName, Stu_SuperLNo)
   MODULE(Code, Descr, Credit)
   PREREQU(Code, PCode, MinMark)
   TAKES(Stu_No, Code, Year, Mark)
   TEACH(L_No, Code, Semester)

The table below explains the attributes used in the database:

Lecturer		Student	
L_No,	Unique lecturer number	Stu_No,	Unique student number
L_FName,	First name	Stu_FName,	First name
L_LName,	Last name	Stu_LName,	Last name
L_Office,	Office number	Stu_SuperLNo	Supervisor (lecturer) number of lecturer that supervises student
DName,	Department where lecturer teaches		
L_email	Email		
Module		**Department**	
Code,	Module code (e.g. WRDV301)	DName,	Department name
Descr,	Description	HoDLno,	Head of Department unique lecturer number
Credit	Credit value of module (e.g. 7)	Building	Building where department is
Takes		**Prerequ**	
Stu_No,	Student number	Code,	Module code
Code,	Module code	PCode,	Code of prerequisite module
Year,	Year of registration for module	MinMark	Minimum mark required for PCode
Mark	Mark obtained		
Teach			
L_No,	Lecturer teaching the module		
Code,	Module code		
Semester	Semester in which module is taught		

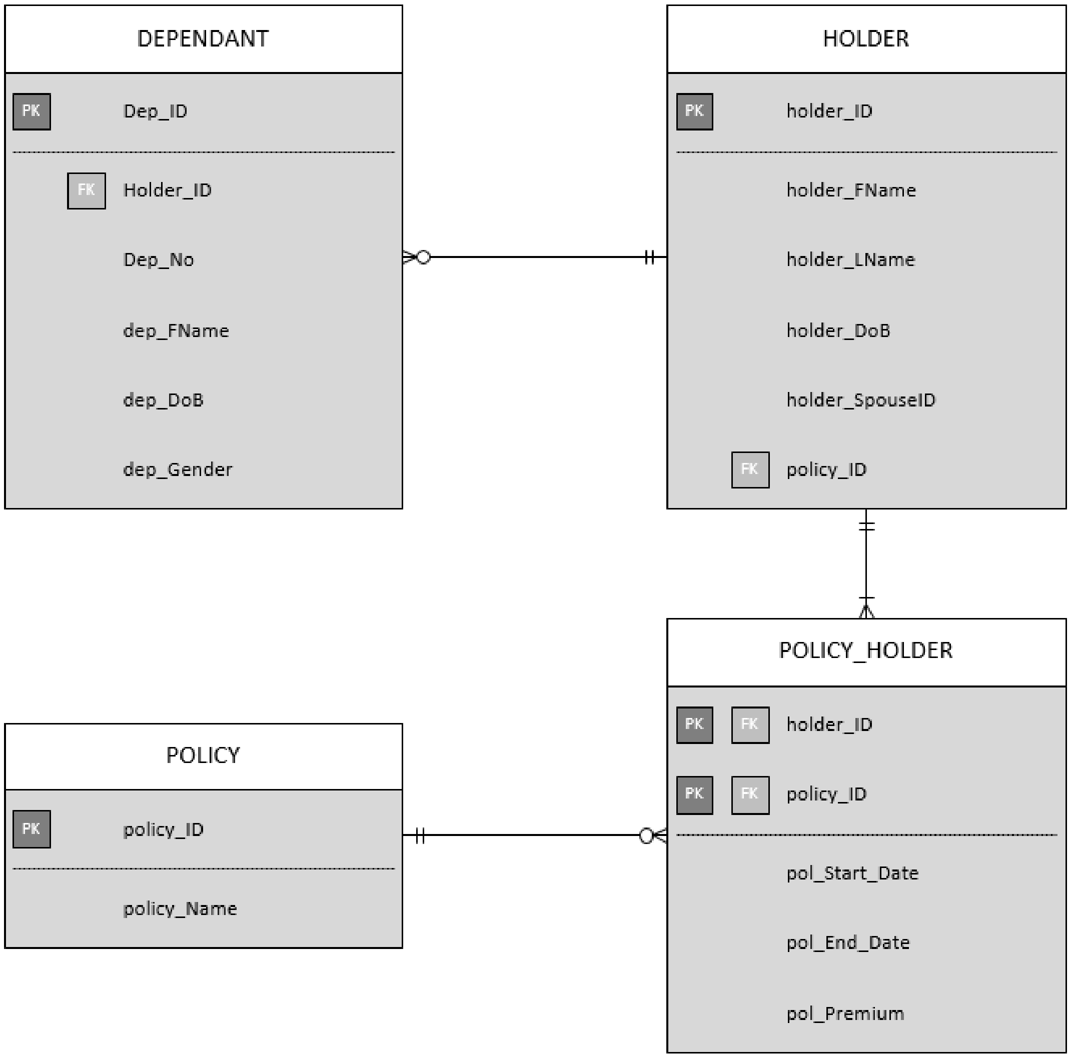
Make use of the database provided (see instructions above) and then create the SQL code that will answer the following queries.

a) Write the SQL to display all module codes with a credit value that is higher than the credit value of all the courses lectured by Dieter Vogts. Your resultant table should include the module code and the lecturer name, using the alias LECTURER. The lecturer name should be displayed in the format l_fname l_lname, e.g. Lynette Barnard. 6

b) Write the SQL code to list the first and last names of the heads of departments and the number of students each one of them supervises. Only list the names for heads of department who supervise at least three students. 4

c) Write the SQL code to return a list of all the first and last names of each lecturer as well as the first and last names of each lecturer's head of department. Use the following aliases for the first and last names of the heads of department: HoDFName and HoDLName. Order the results in descending order of head of department last name. 4

7. Refer to the model provided below when answering question 6. This database stores policy, policy holder and policy holder dependant information. The average policy premium paid by policy holders is R400. In exceptional cases the policy premium will be less than R50. Policy holders can have multiple dependants and each dependant has a dependant number (e.g. 1, 2, 3). A maximum of five dependants is allowed per policy holder. About 85% of all current policy holders have a maximum of 2 dependants. [6]



Rewrite the following query to make it more efficient. Identify the changes you have made and explain why your changes will make the query more efficient. You need only rewrite the changes you have made.

```
SELECT      policy_holder.policy_id, holder.policy_id, holder_fname, holder_lname,
            dep_id, dep_fname
FROM        holder, dependant, policy_holder
WHERE       holder.holder_id=dependant.holder_id and
            holder.policy_id=policy_holder.policy_id and
            NOT (dep_no<=3) and
            holder_dob<#31/12/1940# and
            pol_premium < 50;
```

TOTAL 100

Explanation of information on the invoice

- The Company details at the top of the form need not be included in your database design.
- SOLD TO: This section includes information of the customer that had placed the order.
- DELIVER TO: This section includes the name and address details of where the ordered items need to be delivered. The delivery may be made to the customer (whose details are in the SOLD TO area), or it may be made to someone else that is not a customer.
- PO No: this is the Purchase Order number the customer needs to provide to place an order. A purchase order is associated with an order number (see Our Order No below).
- Our Order No: this is the order number given by the company for the order placed. An order number is linked to one PO No (see PO No above). An order will have a specific order date.
- An order is recorded on an invoice (the document you have) and an order may contain many items that are ordered.

INVOICE

Invoice date: 12 May 2019
Invoice # 10125

LION BRAND
229 Govan Mbeki Ave
Central
Port Elizabeth, 6001
041 458 74545
sales@holden.co.za

SOLD TO Smith Rentals
57 Beetlestone Road
Gelvandale, 6020
084 7897 895
Customer No SM2304

DELIVER TO JBL Smith
122 High Street
Grahamstown, 6148
073 5789 221

PO NO	OUR ORDER NO	ORDER DATE	DELIVERY DATE	SALES PERSON
PO3351	12124	9 April 2019	14 May 2019	10 (Kumalo, Xola)

QTY	ITEM	DESCRIPTION	UNIT PRICE (EXCL)	DISCOUNT	LINE TOTAL
5	AT414	High-back office chair	1255.00	0.00	6275.00
7	BZ287	Office desk – oval	2500.00	0.00	17500.00
3	DT202	Printer table	899.00	0.00	2697.00
Total Discount				0.00	
				SUBTOTAL	26472.00
				VAT	3970.80
				DELIVERY	350.00
				TOTAL	30792.80

Thank you for your business!